

Traffic Monitoring System

<https://github.com/aiyshwary/Traffic-Monitoring-System>

PHP MySQL JavaScript HTML CSS XAMPP

OVERVIEW

A PHP and SQL-based web application that supports traffic controllers with real-time violator tracking, periodic traffic status notifications, daily reports, and transport department recommendations for infrastructure improvements.

IMPACT

Developed a multi-role Traffic Monitoring System web application with separate portals for admins, officers, and citizens, enabling automated violator tracking, fine management, road signalling status, and daily traffic reporting.

TECHNICAL WALKTHROUGH

Traffic Monitoring System

(TMS)

Date: March 8, 2026

Built with PHP, MySQL, Apache (MAMP) | Bootstrap 4 Frontend

Project Overview

The Traffic Monitoring System (TMS) is a web-based application designed to streamline traffic management for both authorities (Officers) and the public (Citizens). It provides tools for monitoring road conditions, managing traffic signal timings, and handling traffic violation fines - including online payment. The system is built using PHP for server-side logic, MySQL as the relational database, Apache as the web server (via MAMP), and Bootstrap 4 for a responsive frontend UI.

Technology Stack

Layer Technology / Library

Backend Language PHP 8.3 (server-side scripting)

Database MySQL 8 (via MySQLi extension)

Web Server Apache (MAMP on macOS)

Frontend Framework Bootstrap 4

JavaScript Libraries jQuery 2/3, Owl Carousel, ScrollReveal

CSS Libraries Font Awesome 4.7, Animate.css, Perfect Scrollbar

UI Utilities Select2, Tilt.js, Lightbox

Session Management PHP Native Sessions

User Roles

The system has three distinct user roles, each identified by a file prefix convention

i) Prefix Role Access Level

ii) a Admin / Public Landing page, login/register chooser. No authentication required. c Citizen Registered vehicle owners. Can view own fines, pay fines, view signals and roads. o Officer Traffic authority personnel. Can manage fines, edit signal timings, view road data.

iii) File Structure & Purpose

iv) 4.1 Shared / Core Files

v) File Purpose

vi) aconnection.php MySQL DB connection - included by all PHP pages. Holds host, user, pass, dbname. logout.php Destroys the PHP session and redirects the user back to ahome.php. database.sql Full MySQL schema - creates all 5 tables and inserts sample road/signal data. 4.2 Public / Admin Pages (prefix: a)

vii) File Purpose

viii) ahome.php Public homepage. Slider, About, Features, Team, Contact sections. alioption.php Login chooser page - two buttons: Citizen Login or Officer Login. asuooption.php Register chooser page - two buttons: Citizen Signup or Officer Signup. alogout.php Legacy logout file kept for compatibility.

ix) 4.3 Citizen Pages (prefix: c)

x) File Purpose

xi) csignup.php Registration form. Inserts into 'citizen' table AND auto-creates a fine record. clogin.php Login form. Validates against 'citizen' table, sets \$_SESSION['email']. cfunctions.php check_login() function - validates session, redirects to clogin.php if absent. chome.php Citizen dashboard with slider linking to all citizen features. cfine.php Shows own fine records (amount, status). Pay Now button if unpaid. cpayment.php Card payment UI. On submit sets fine.paid = 1 in database. csignalling.php Read-only view of signal timings (road name, time_from, time_to). croad.php Read-only view of all roads (station name, road name).

xii) 4.4 Officer Pages (prefix: o)

xiii) File Purpose

xiv) signup.php Registration form for officers. Stores username, email, OID, password. ologin.php Login form. Validates against 'officer' table, sets \$_SESSION['email']. ofunctions.php check_login() function - validates session, redirects to ologin.php if absent. ohome.php Officer dashboard with slider linking to all officer features. ofine.php Lists ALL fines in the system with an Edit button per row. ozfupdate.php Edit form to update a specific fine amount by fine ID. osignalling.php Lists all signal timings with an Edit button per row. osupdate.php Edit form to update time_from and time_to for a signal entry. oroad.php View all roads - Road ID, Station Name, Road Name.

xv) Database Schema

xvi) Database name: 'project' | Character set: utf8mb4 | Engine: InnoDB

xvii) Table: citizen - Stores registered citizens

xviii) Column Type Constraints

xix) INT AUTO_INCREMENT, PRIMARY KEY

xx) username VARCHAR(100) NOT NULL

xxi) email VARCHAR(100) NOT NULL, UNIQUE

xxii) phone VARCHAR(15)

xxiii) ID VARCHAR(20) Vehicle ID

xxiv) pass VARCHAR(255) NOT NULL (plain text - upgrade to hashed in production)

xxv) Table: officer - Stores traffic officers

xxvi) Column Type Constraints

xxvii) INT AUTO_INCREMENT, PRIMARY KEY

xxviii) username VARCHAR(100) NOT NULL

xxix) email VARCHAR(100) NOT NULL, UNIQUE

xxoID VARCHAR(20) Officer ID

xxpiass VARCHAR(255) NOT NULL

xxTable: fine - Traffic violation fines

xxColumn Type Constraints

xxidINT AUTO_INCREMENT, PRIMARY KEY

xxusername VARCHAR(100) Links to citizen.username

xxVID VARCHAR(20) Vehicle ID

xxamount DECIMAL(10,2) Default 0.00

xxpaid TINYINT 0 = Pending, 1 = Paid

xxTable: road - Road information

xlColumn Type Constraints

xliid INT AUTO_INCREMENT, PRIMARY KEY

xliisstation_name VARCHAR(100)

xliiroad_name VARCHAR(100)

xliiTable: signalling - Traffic signal timings per road

xliiColumn Type Constraints

xliiID INT AUTO_INCREMENT, PRIMARY KEY

xliiroad_name VARCHAR(100)

xlvi) time_from VARCHAR(20) e.g. 06:00

lix) time_to VARCHAR(20) e.g. 10:00

I) Complete Application Flow

li) 6.1 Citizen Flow

lii) Step Page Action

liii) Step 1 Visit ahome.php Public homepage. Click 'Register Now'.

liv) Step 2 asuoption.php Choose 'CITIZEN' to go to csignup.php.

lv) Step 3 csignup.php (POST) Fill name, email, phone, vehicle ID, password. Two DB inserts: citizen table + fine table (amount=0).

lvi) Step 4 clogin.php (POST) Enter email + password. Session set on success. Redirect to chome.php.

lvii) Step 5 chome.php Dashboard. Links to Fine Payment, Signal Timings, Google Maps, Road Info.

lviii) Step 6 cfine.php View own fine. If amount > 0 and unpaid, 'Pay Now' link appears.

lix) Step 7 cpayment.php Enter card details. On submit: fine.paid = 1 in DB. Redirected back with success message.

lx) Step 8 logout.php Session destroyed. Redirect to ahome.php.

lxi) 6.2 Officer Flow

lxii) Step Page Action

lxiii) Step 1 ahome.php Click 'Register Now'.

lxiv) Step 2 asuoption.php Choose 'OFFICER' to go to osignup.php.

Step 3 osignup.php (POST) Fill name, email, Officer ID, password. Insert into officer table.

Step 4 ologin.php (POST) Login. Session set. Redirect to ohome.php.

Step 5 ohome.php Dashboard. Links to Fine Allotment, Signal Timings, Google Maps, Road Info.

Step 6 ofine.php View ALL citizen fines. Each row has an Edit link.

Step 7 ofupdate.php Edit fine amount for a specific citizen. Updates fine.amount in DB.

Step 8 osignalling.php View all signal timings. Each row has an Edit link.

Step 9 osupdate.php Update time_from and time_to for a road signal.

Step 10 oroad.php View all road records (read-only).

Step 11 logout.php Session destroyed. Redirect to ahome.php.

Authentication & Session Management

Both Citizens and Officers use PHP's native session mechanism

i) Login: `$_SESSION['email'] = $user_data['email'];` (set on successful login) Guard: Every protected page includes `check_login($con)` from the role's functions file.

ii) If the session is not set, the user is immediately redirected to the login page.

iii) Logout: `unset($_SESSION['email']); header('Location: ahome.php');`

Two separate `check_login()` functions exist

i) cfunctions.php - queries the 'citizen' table, redirects to clogin.php if unauthenticated.
ofunctions.php - queries the 'officer' table, redirects to ologin.php if unauthenticated.

ii) Setup & Deployment Guide

iii) Requirements

iv) MAMP (or XAMPP) with Apache + MySQL

v) PHP 8.x

vi) Browser (Safari, Chrome, Firefox)

vii) Installation Steps

viii) Copy project folder to /Applications/MAMP/htdocs/Traffic-Monitoring-System/

ix) Open MAMP and click 'Start Servers' (both Apache and MySQL must be green).

x) Open phpMyAdmin at <http://localhost:8888/phpMyAdmin>

xi) Create a new database named 'project'.

xii) Select 'project' database -> Import tab -> choose database.sql -> click Go.

xiii) Open the site at <http://localhost:8888/Traffic-Monitoring-System/ahome.php>

xiv) Database Credentials (aconnection.php)

xv) Parameter Value

xvi) Host localhost

xvii) Username root

xviii) Password root (MAMP default)

xix) Database project

xx) Security Notes & Known Limitations

xxi) Issue Detail

xxiSQL Injection Queries use raw string interpolation. Use prepared statements (PDO/MySQLi) in production.

xxiiPlain-text passwords Passwords stored as plain text. Use password_hash() and password_verify() in production.

xxiiiNo admin role There is no separate admin login. ahome.php is publicly accessible.

xxivNo input sanitization User input is not sanitised before DB insertion. Add htmlspecialchars/strip_tags.

xxvPayment is simulated cpayment.php marks fine as paid but does not connect to a real payment gateway.

xxviCSRF protection Forms have no CSRF tokens. Add tokens to all POST forms in production.

xxviiIntend Assets & Directory Layout

xxviiiDirectory Contents

xxixassets/css/ Bootstrap, Font Awesome, Owl Carousel, Lightbox, main template CSS assets/js/ jQuery, Bootstrap JS, Owl Carousel, ScrollReveal, Lightbox, custom.js assets/images/ Slider images (a.jpg, b.jpg, c.jpg), icons, team photos

xxxvendor/bootstrap/ Bootstrap 4 CSS + JS

xxxivendor/animate/ Animate.css for scroll animations

xxxiivendor/select2/ Select2 dropdown library

xxxiiivendor/perfect-scrollbar/ Smooth scrollbar for tables vendor/tilt/ Tilt.js - 3D tilt effect on login images vendor/css-hamburgers/ Hamburger menu animation CSS css/ util.css + main.css - login page layout styles

xxxivfont-awesome-4.7.0/ Icon font library

xxxvimages/icons/ Favicon and SVG illustrations (citizen.svg, officer.svg, traffic.svg)

KEY DETAILS

1. Admin portal manages user accounts, officer assignments, road data, and signalling configurations.
2. Officer portal logs violators, records fines, and updates real-time road and signal status.
3. Citizen portal supports account registration, violation history lookup, and online fine payment.
4. Automated daily reports summarize traffic violations and flag roads needing infrastructure attention.
5. Built with PHP, MySQL (XAMPP), JavaScript, and CSS with a responsive multi-role front-end.